Universität Stuttgart
Master of Science

# Characterisation of a Distributed Systems

Marco Aiello

## Distributed System Definition

from Coulouris et al.

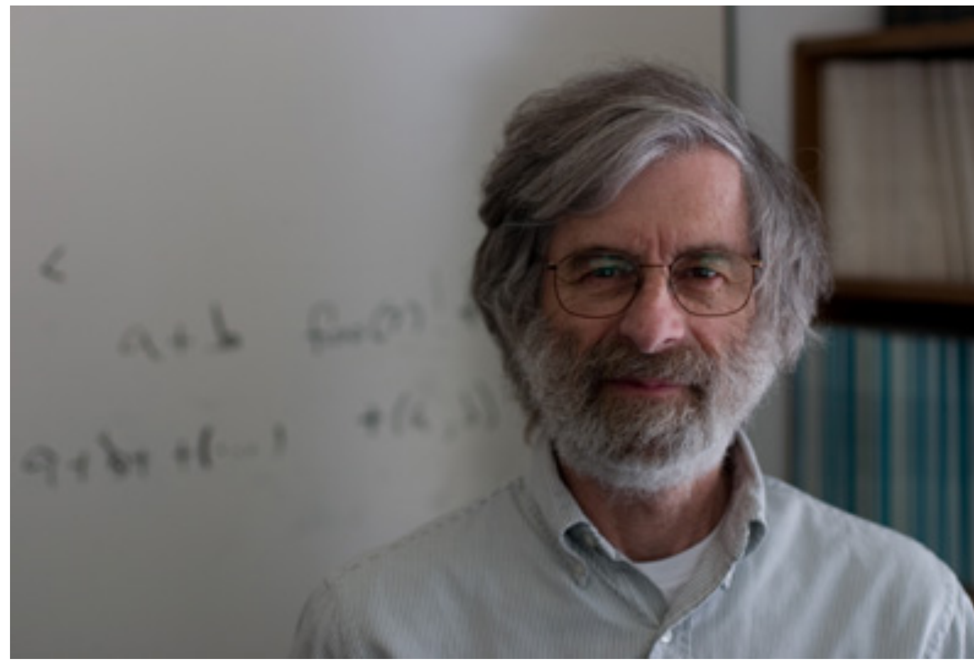
A distributed system is a collection of autonomous hosts that are connected through a computer network. Each host executes computations and operates a distribution middleware, which enables the components to coordinate their activities via message-passing in such a way that users perceive the system as a single, integrated computing facility.

# or better said

# or better said

*“A distributed system is one in which the failure of a machine you have never heard of can cause your own machine to become unusable”*

Leslie Lamport

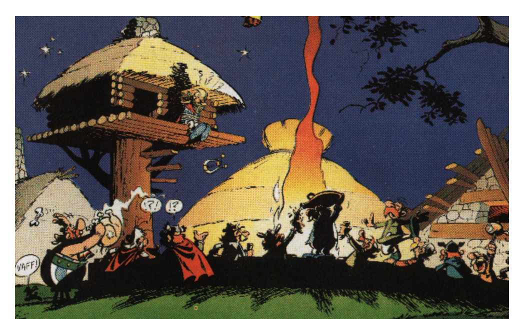


# How to think in Distributed Systems Terms

- Verify correctness of protocols
- Verify efficiency of protocols
- Hosts are independent state machines
- Deal with concurrency, consistency, replication

# Two generals problem

- Two Generals need to coordinate an attack against an enemy. If they attack individually, they will loose, if they attack together they will win.
- But the enemy lies in the middle and can intercept the coordination messages and avoid delivery
- Can the generals defeat the enemy?



# Two generals problem

- Two Generals need to coordinate an attack against an enemy. If they attack individually, they will loose, if they attack together they will win.
- But the enemy lies in the middle and can intercept the coordination messages and avoid delivery
- Can the generals defeat the enemy?




# Two generals problem

## Proof sketch

- **Theorem:** there is no non-trivial protocol that guarantees that the generals will always attack simultaneously

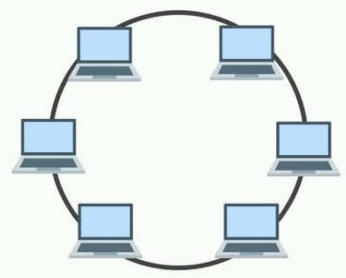
- **Proof:** Ab absurdum, suppose there is one such protocol that does the job in the minimum number of steps n>0.
- Consider the last message sent, the n-th. The state of the sender cannot depend on its receipt, the state of the receiver cannot depend on its arrival, so they both do not need the n-th message. So we would have a protocol with n-1 messages. But that contradict the hypothesis

- **Fact:** A solution requires reliable message delivery.

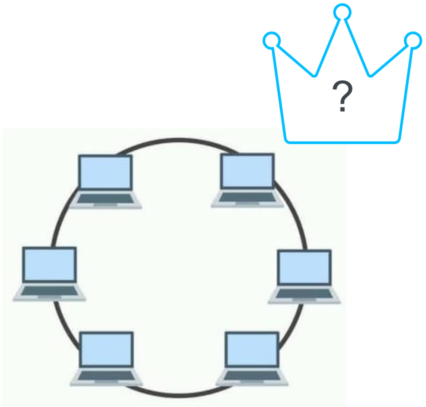
## Some important (negative) results

- No leader election in anonymous rings
- No consensus in asynchronous systems
- Byzantine fault tolerance with at most $\lceil \frac{1}{3}n - 1 \rceil$
- CAP theorem  (Consistency, Availability, Partitioning Tolerance)

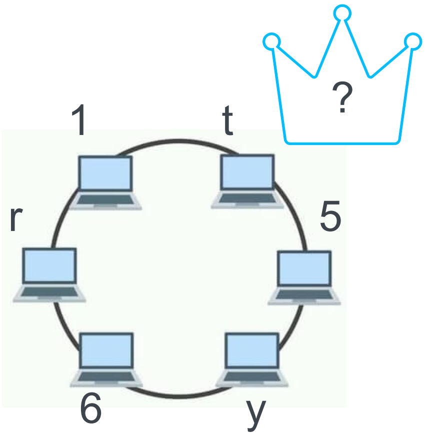
# In pictures



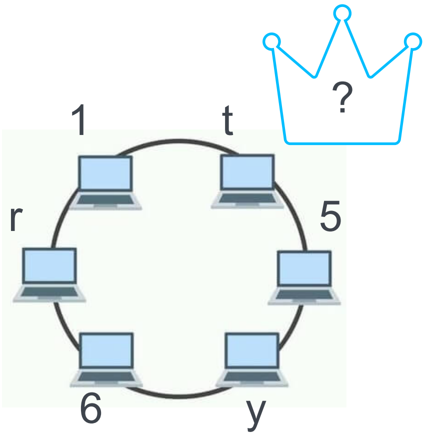
# In pictures

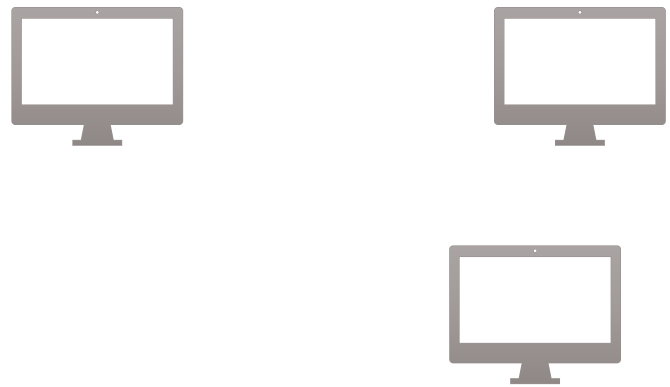


# In pictures

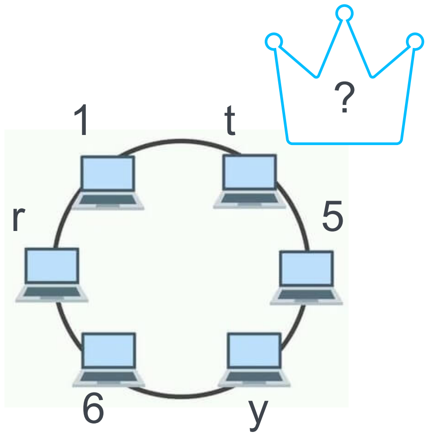


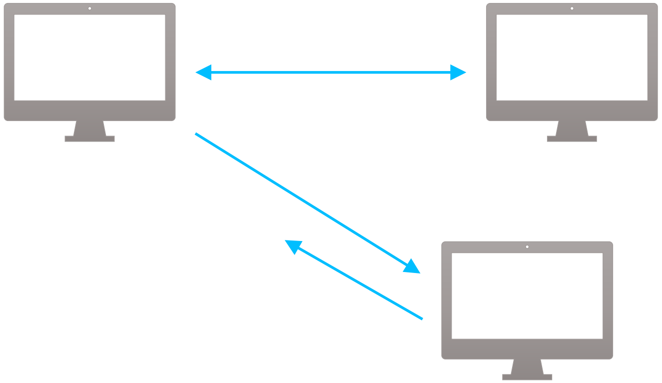
# In pictures



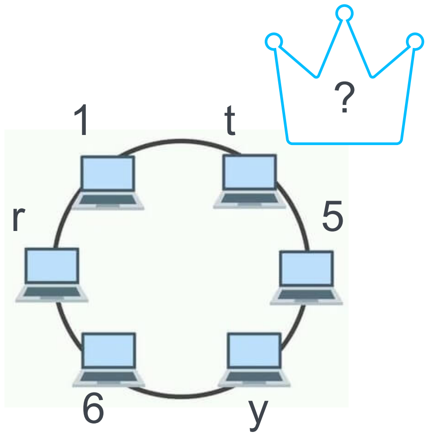



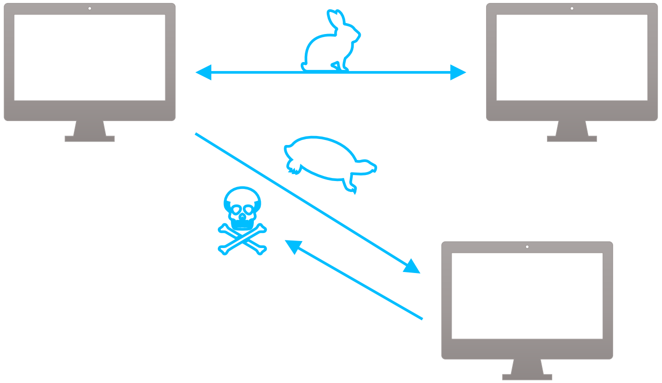
# In pictures



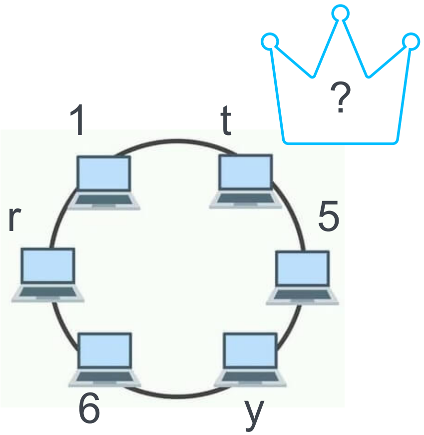



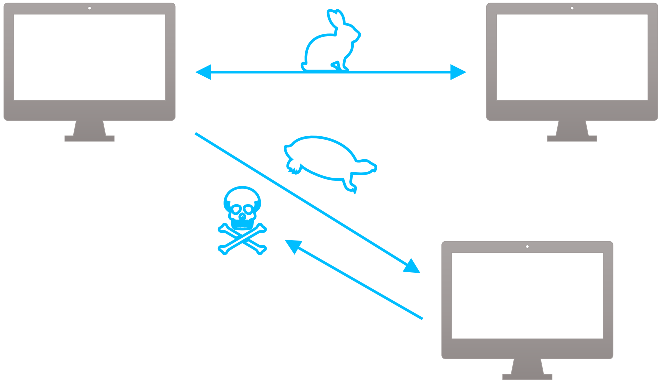
# In pictures

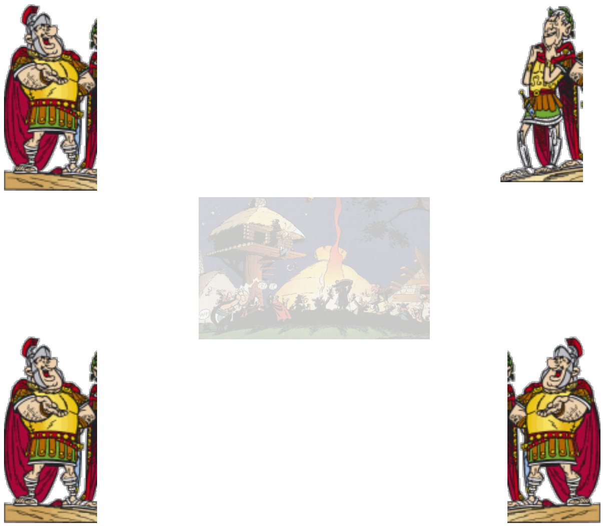


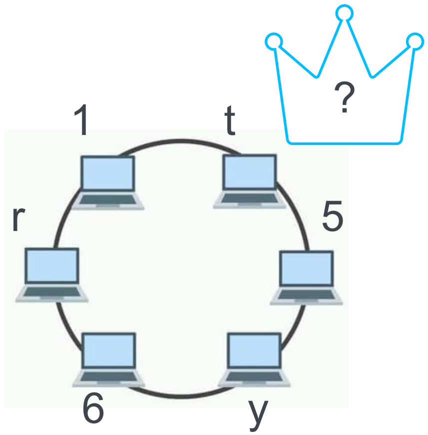
# In pictures

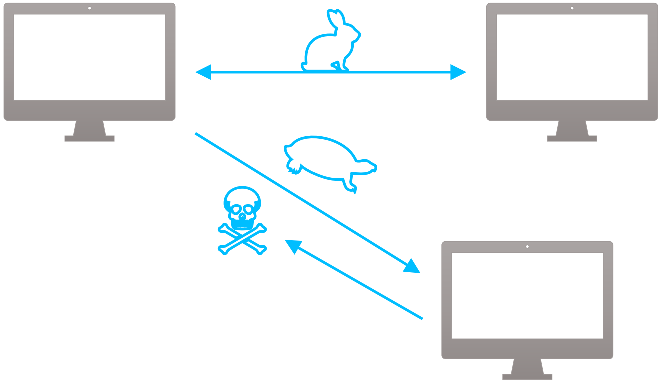


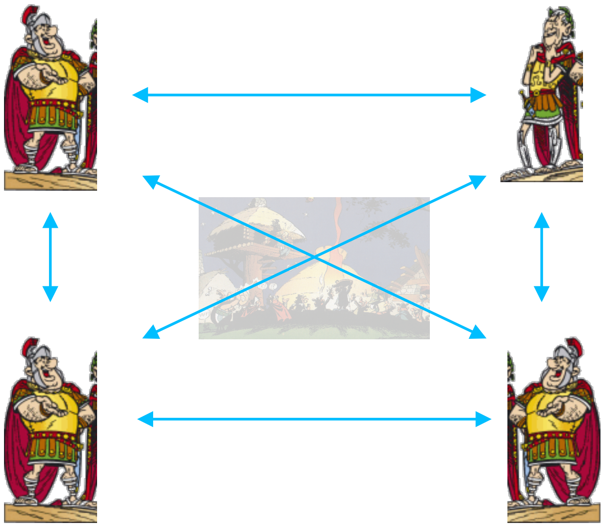





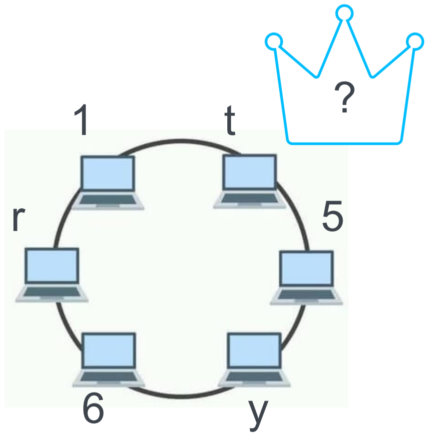
# In pictures

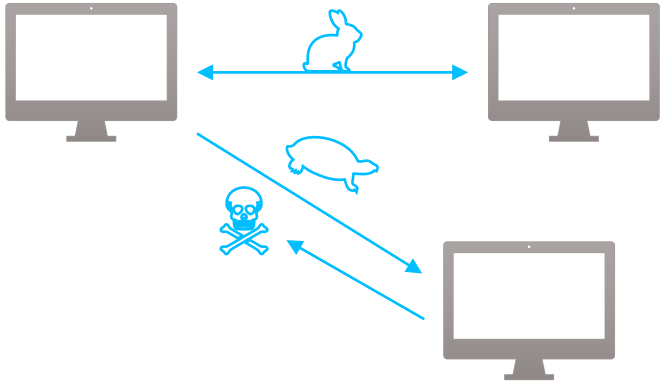


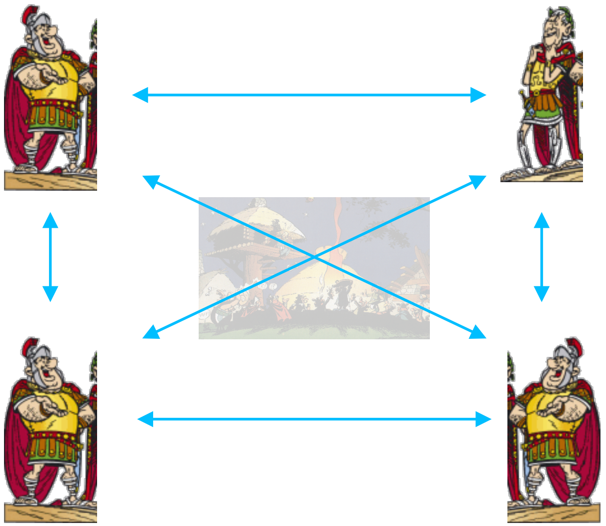



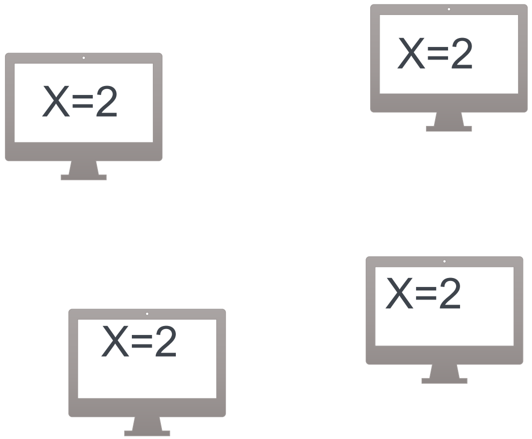


# In pictures

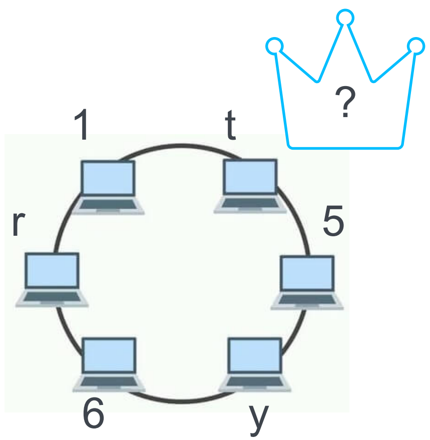


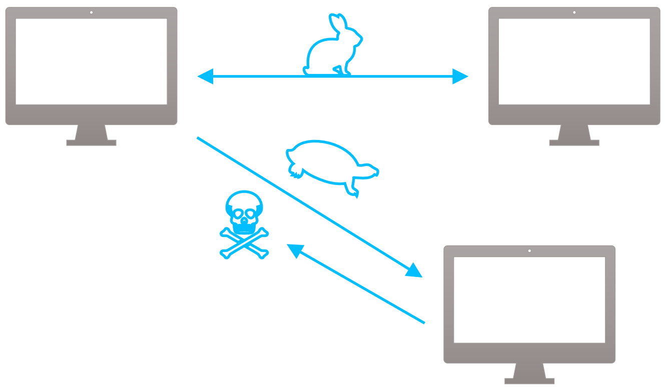



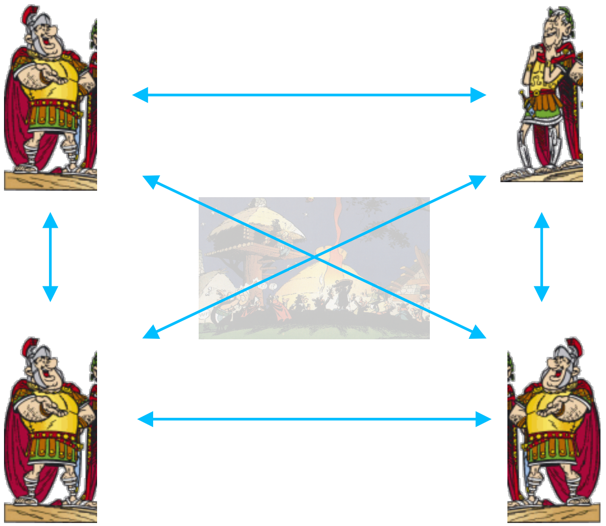




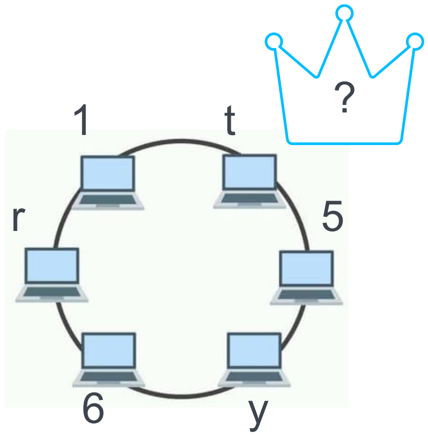
# In pictures

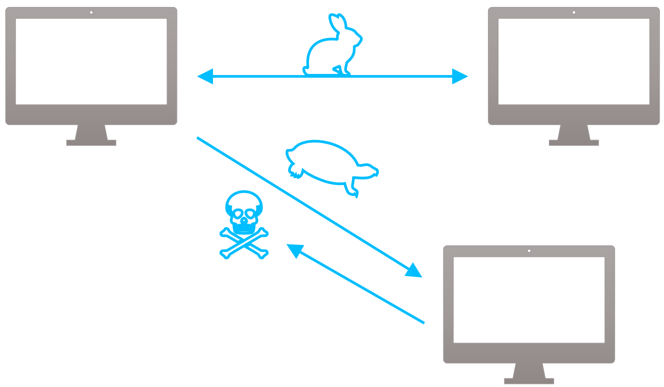


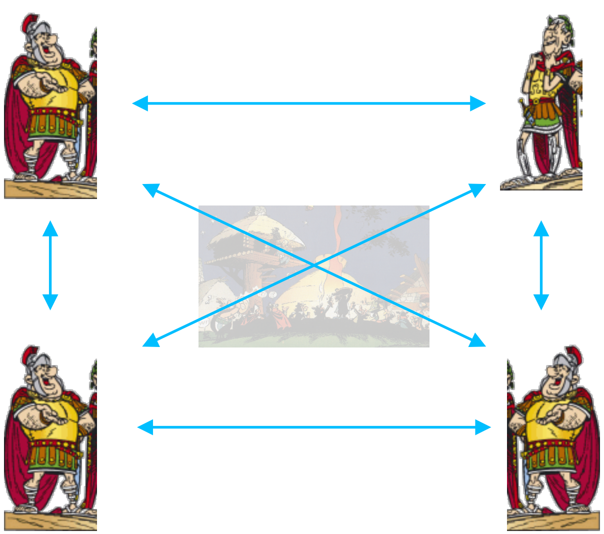





# In pictures



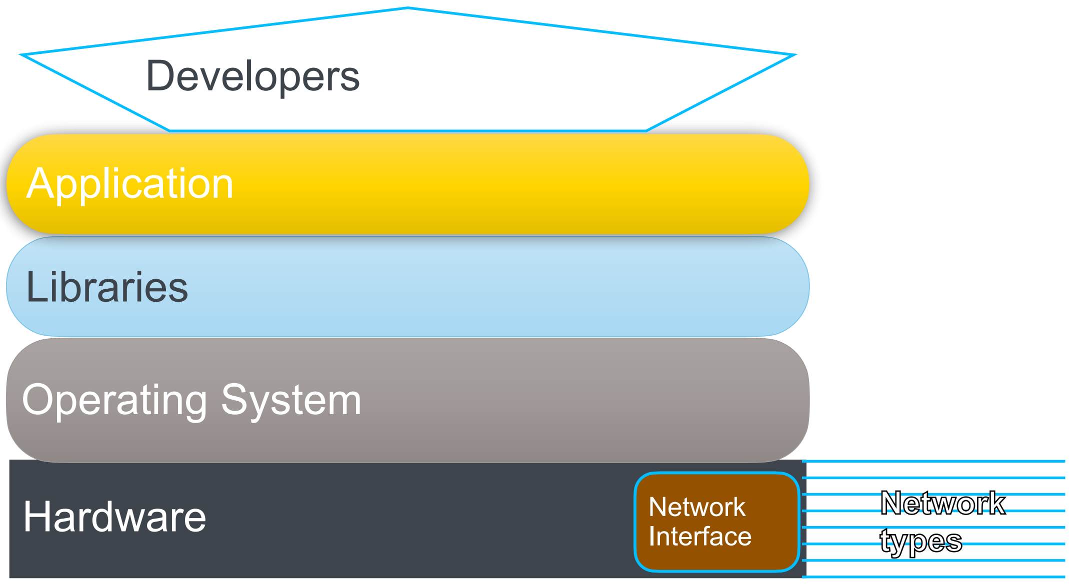


# Distributed Systems Challenges

- Heterogeneity
- Openness
- Security
- Scalability
- Failure handling
- Consistency
- Concurrency
- Transparency

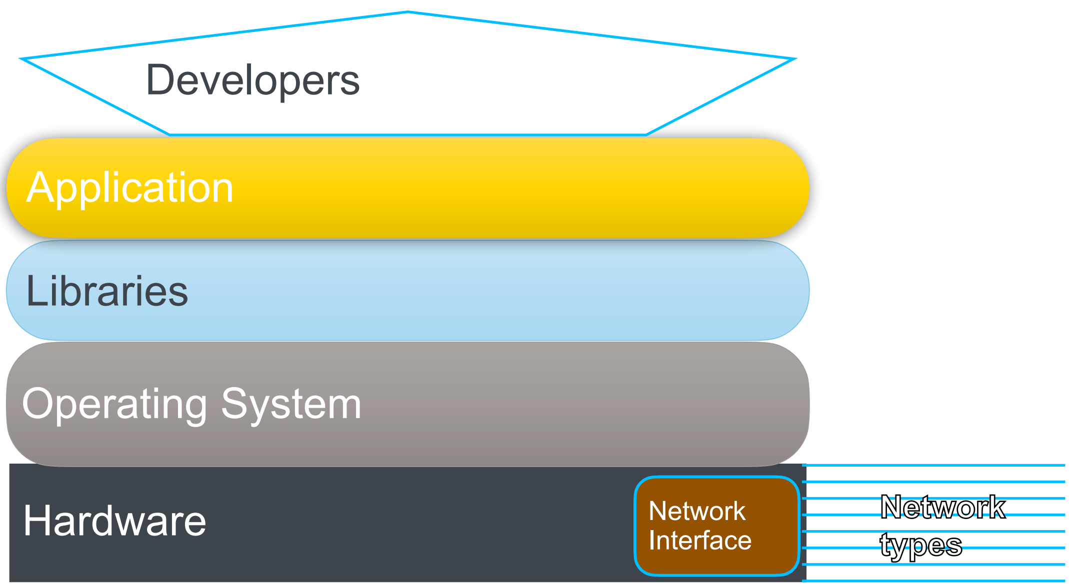
# Distributed Systems Challenges

- Heterogeneity
- Openness
- Security
- Scalability
- Failure handling
- Consistency
- Concurrency
- Transparency



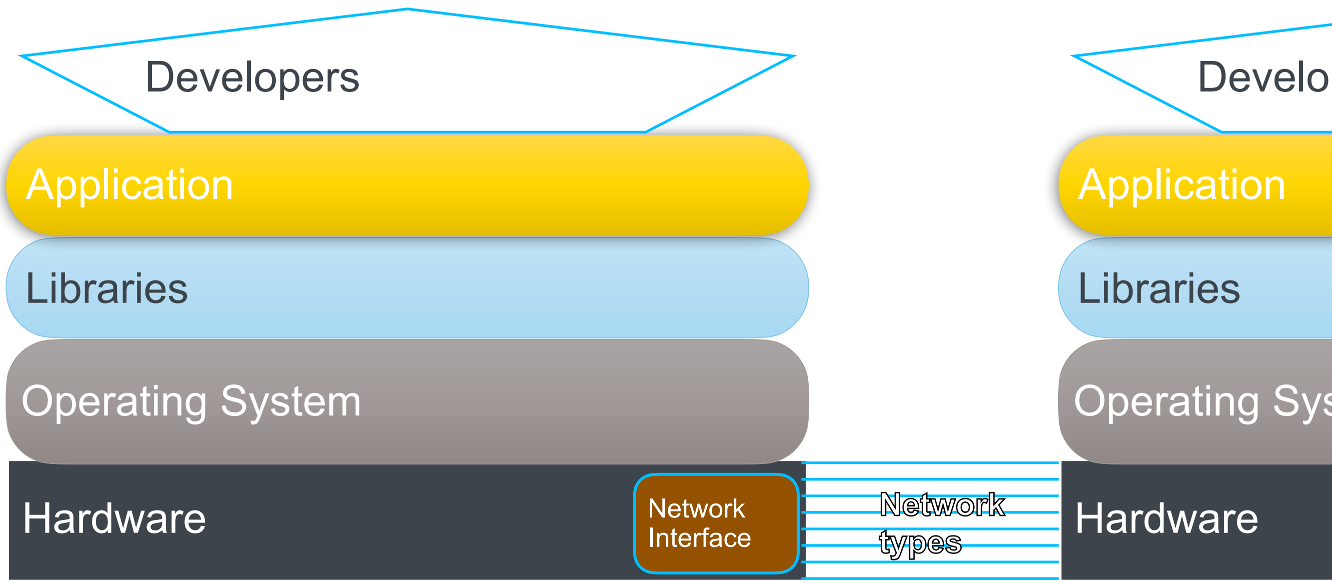
# Distributed Systems Challenges

- Heterogeneity
- Openness
- Security
- Scalability
- Failure handling
- Consistency
- Concurrency
- Transparency



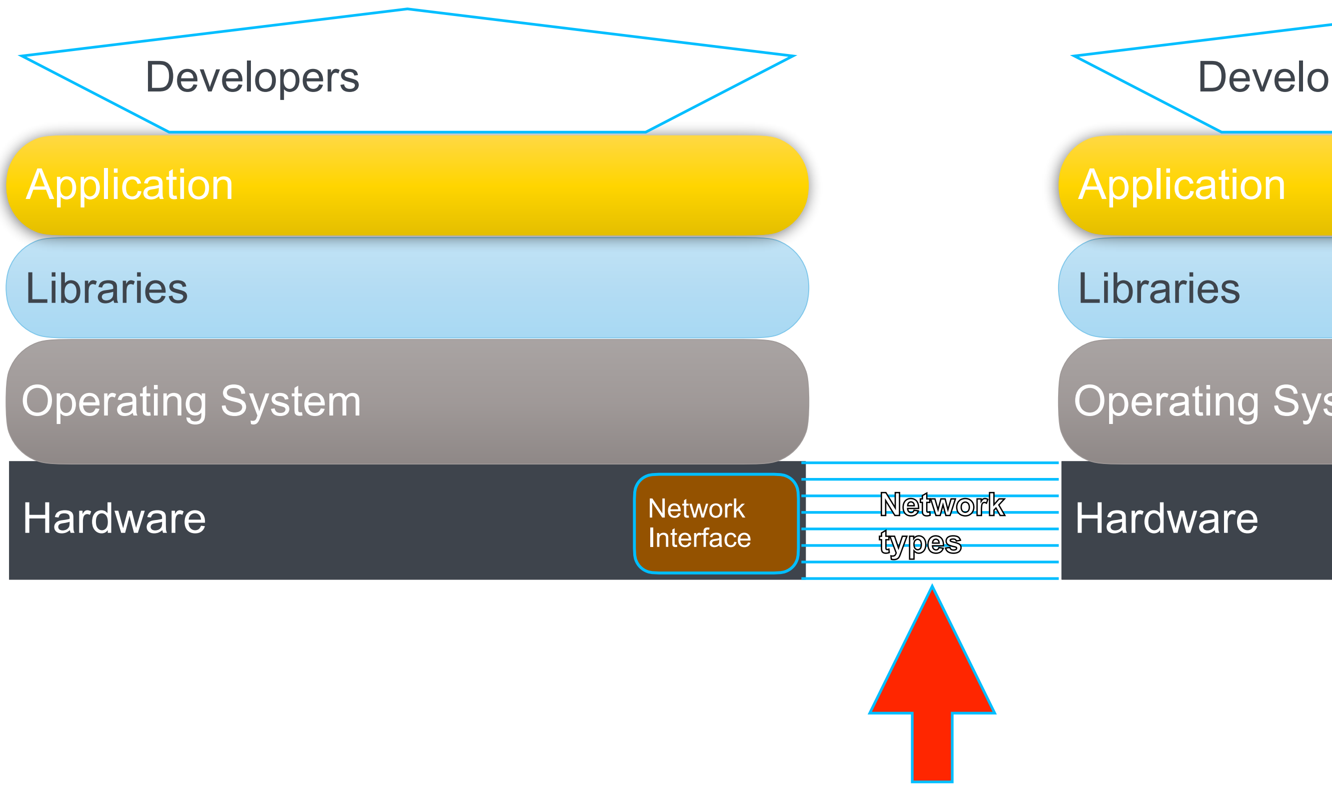
# Distributed Systems Challenges

- Heterogeneity
- Openness
- Security
- Scalability
- Failure handling
- Consistency
- Concurrency
- Transparency



# Distributed Systems Challenges

- Heterogeneity
- Openness
- Security
- Scalability
- Failure handling
- Consistency
- Concurrency
- Transparency

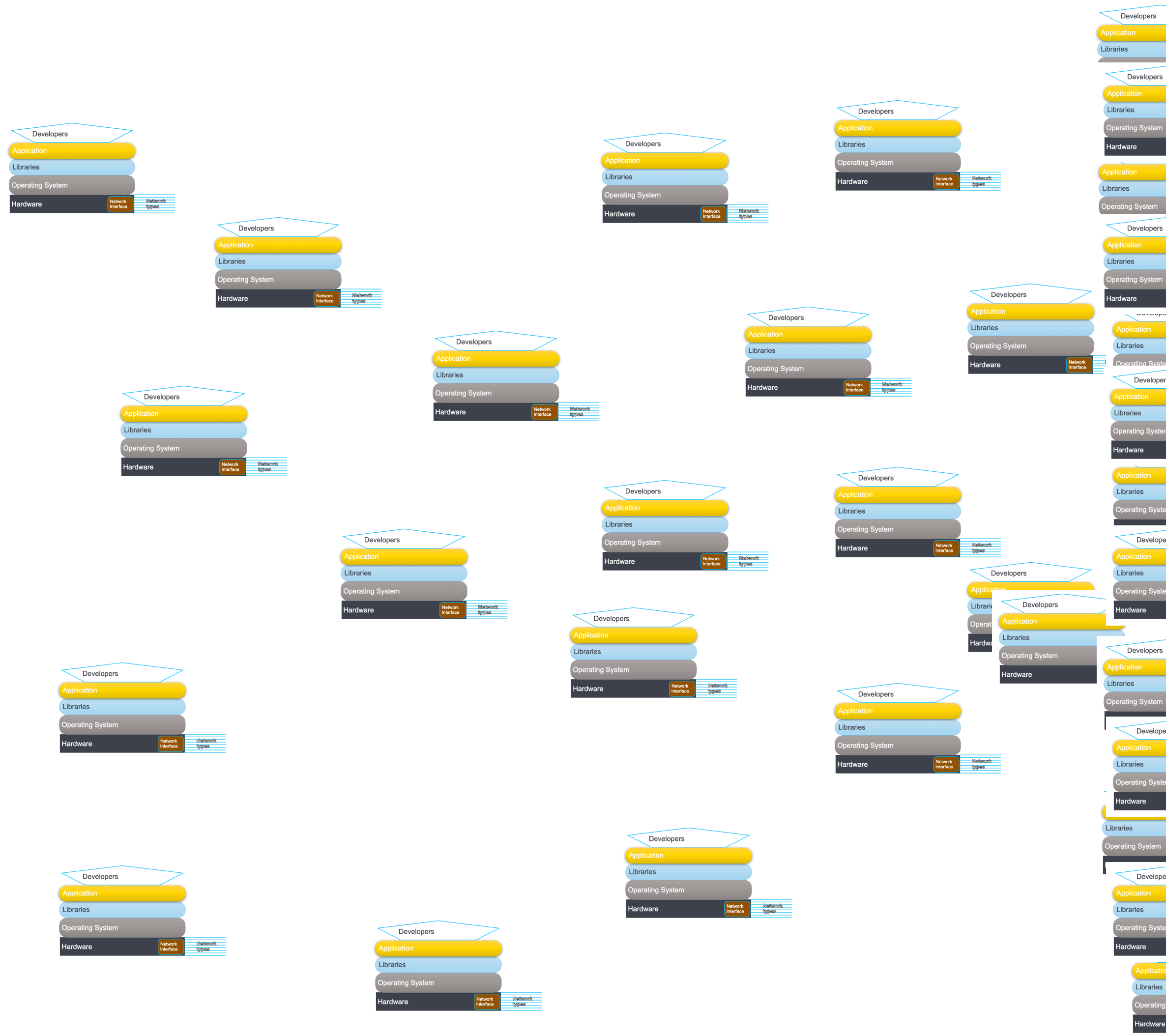


# Distributed Systems Challenges

- Heterogeneity
- Openness
- Security
- Scalability
- Failure handling
- Consistency
- Concurrency
- Transparency

# Distributed Systems Challenges

- Heterogeneity
- Openness
- Security
- Scalability
- Failure handling
- Consistency
- Concurrency
- Transparency

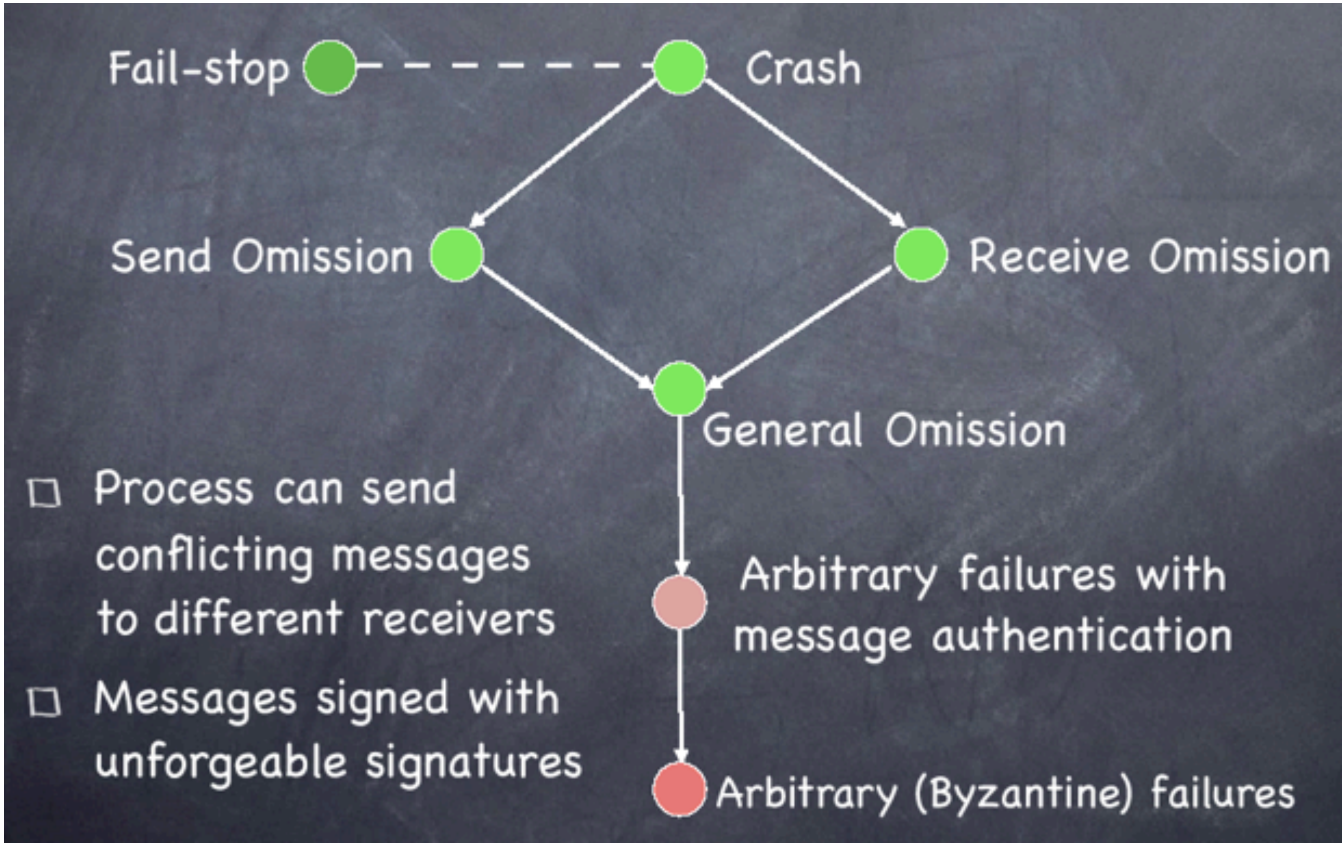


# Distributed Systems Challenges

- Heterogeneity
- Openness
- Security
- Scalability
- Failure handling
- Consistency
- Concurrency
- Transparency

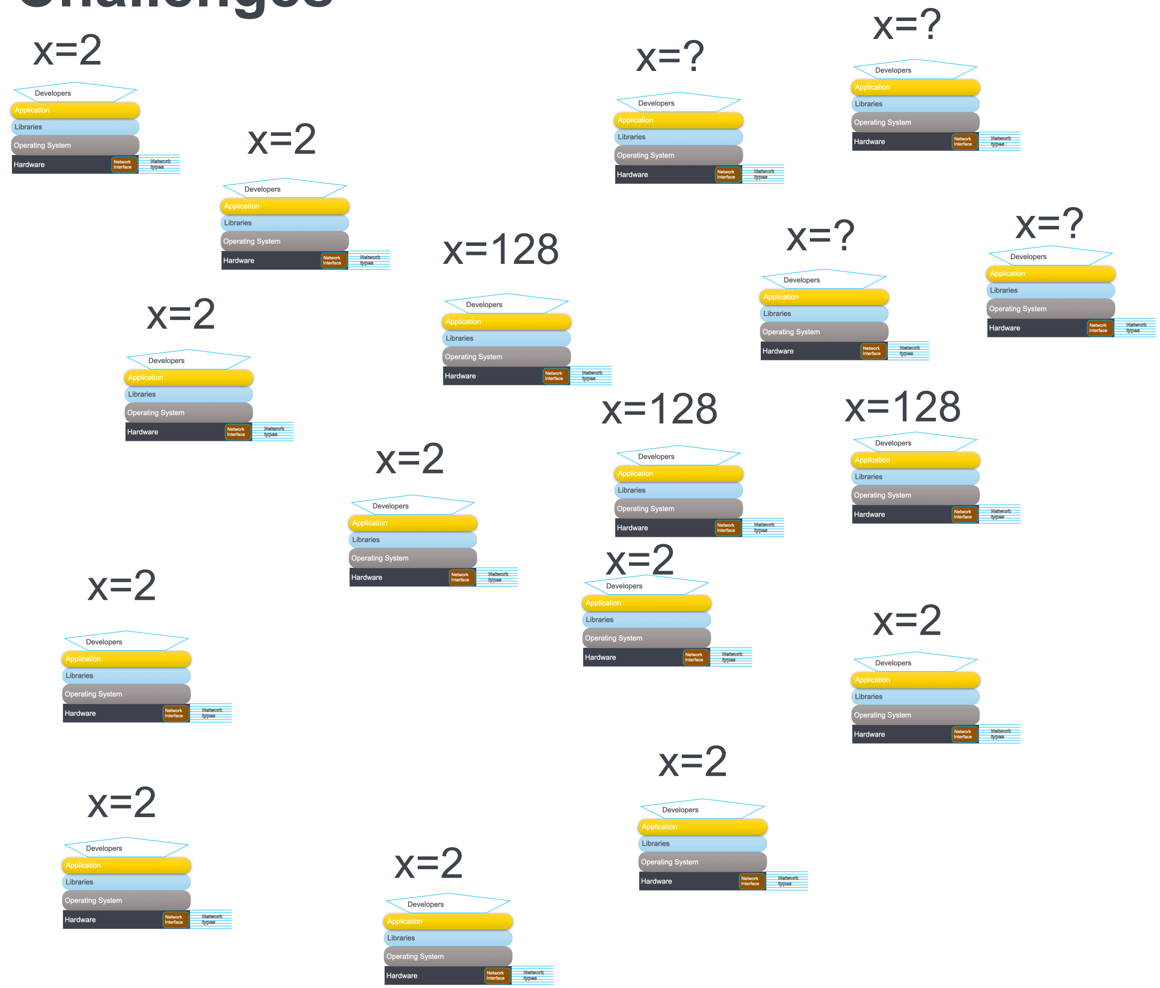
# Distributed Systems Challenges

- Heterogeneity
- Openness
- Security
- Scalability
- Failure handling
- Consistency
- Concurrency
- Transparency



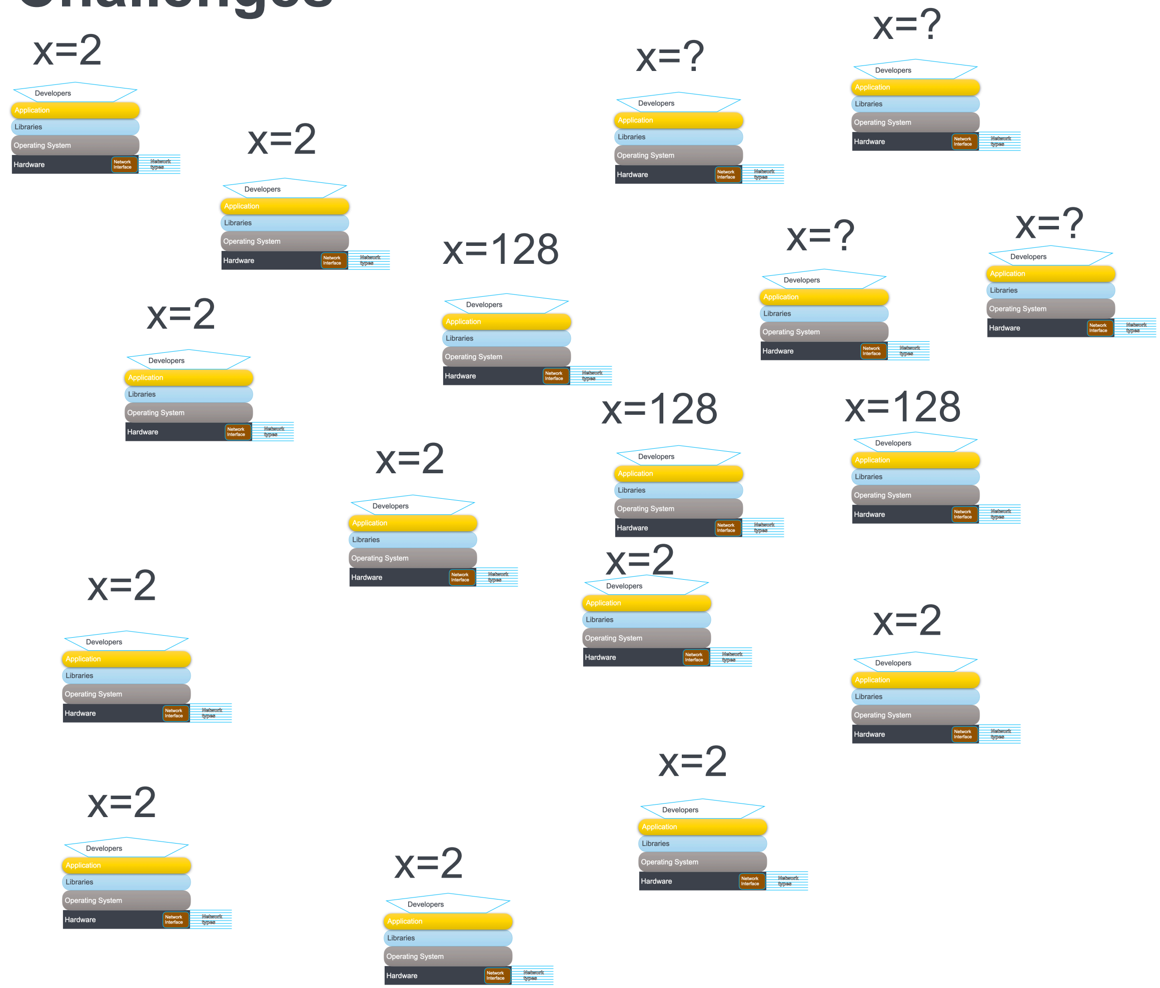
# Distributed Systems Challenges

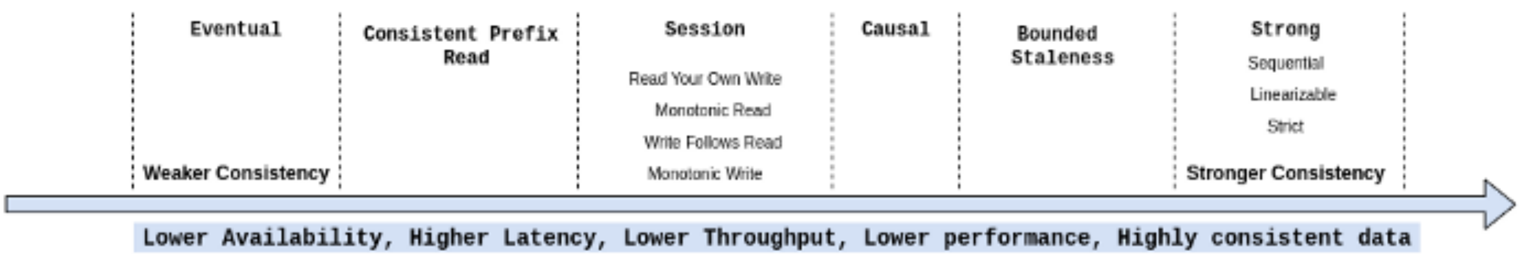
- Heterogeneity
- Openness
- Security
- Scalability
- Failure handling
- Consistency
- Concurrency
- Transparency



# Distributed Systems Challenges

- Heterogeneity
- Openness
- Security
- Scalability
- Failure handling
- Consistency
- Concurrency
- Transparency



# Distributed Systems Challenges

- Heterogeneity
- Openness
- Security
- Scalability
- Failure handling
- Consistency
- Concurrency
- Transparency

```
Program 1

x=y=0
y=y+1
print x,y
```

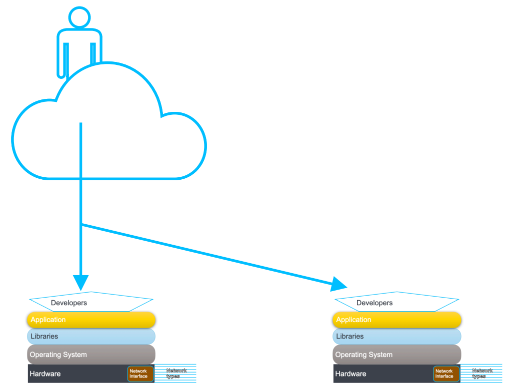
```
Program 2

x=y=3
x=x+1
print x,y
```

```
>run Program 1
> 0,1
> 0,4
> 3,4
> 4,1
> 4,4
```

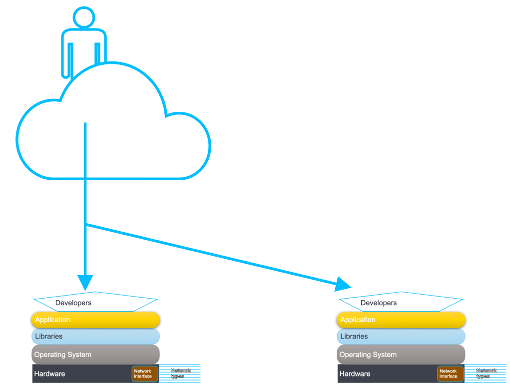
# Distributed Systems Challenges

- Heterogeneity
- Openness
- Security
- Scalability
- Failure handling
- Consistency
- Concurrency
- Transparency
  - Access: *same operations*
  - Location: *any position*
  - Mobility: *dynamic*
  - Replication: *any copy*
  - Failure: *tolerance*
  - Scaling: *any number of nodes*
  - …



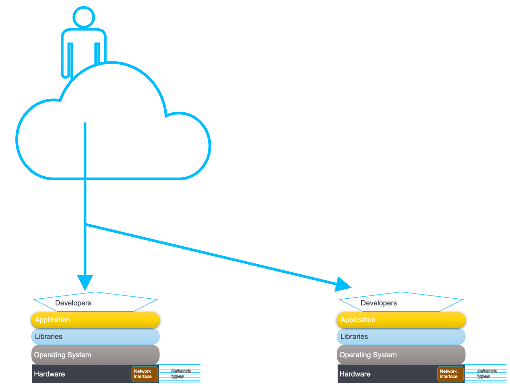
# Distributed Systems Challenges

- Heterogeneity
- Openness
- Security
- Scalability
- Failure handling
- Consistency
- Concurrency
- Transparency
  - Access: *same operations*
  - Location: *any position*
  - Mobility: *dynamic*
  - Replication: *any copy*
  - Failure: *tolerance*
  - Scaling: *any number of nodes*
  - …



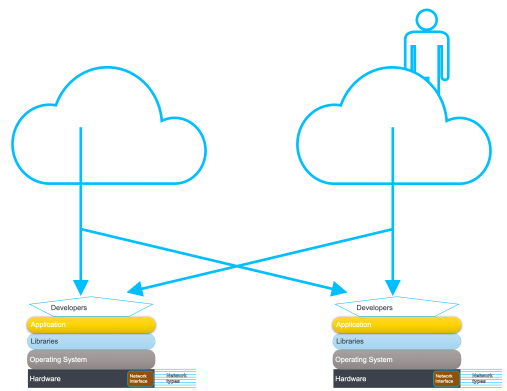
# Distributed Systems Challenges

- Heterogeneity
- Openness
- Security
- Scalability
- Failure handling
- Consistency
- Concurrency
- Transparency
  - Access: *same operations*
  - Location: *any position*
  - Mobility: *dynamic*
  - Replication: *any copy*
  - Failure: *tolerance*
  - Scaling: *any number of nodes*
  - …



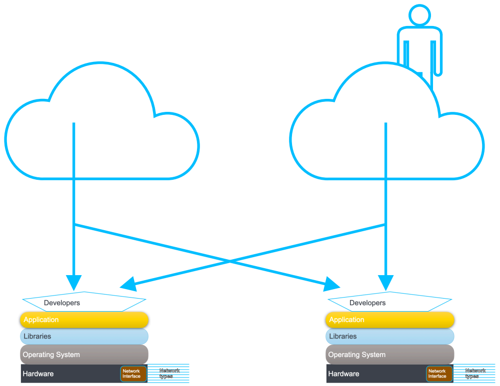
# Distributed Systems Challenges

- Heterogeneity
- Openness
- Security
- Scalability
- Failure handling
- Consistency
- Concurrency
- Transparency
    - Access: *same operations*
    - Location: *any position*
    - Mobility: *dynamic*
    - Replication: *any copy*
    - Failure: *tolerance*
    - Scaling: *any number of nodes*
    - …



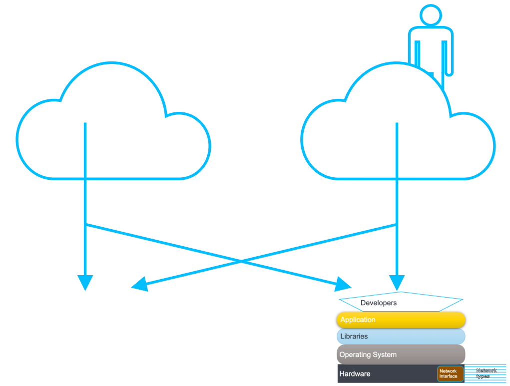
# Distributed Systems Challenges

- Heterogeneity
- Openness
- Security
- Scalability
- Failure handling
- Consistency
- Concurrency
- Transparency
    - Access: *same operations*
    - Location: *any position*
    - Mobility: *dynamic*
    - Replication: *any copy*
    - Failure: *tolerance*
    - Scaling: *any number of nodes*
    - …



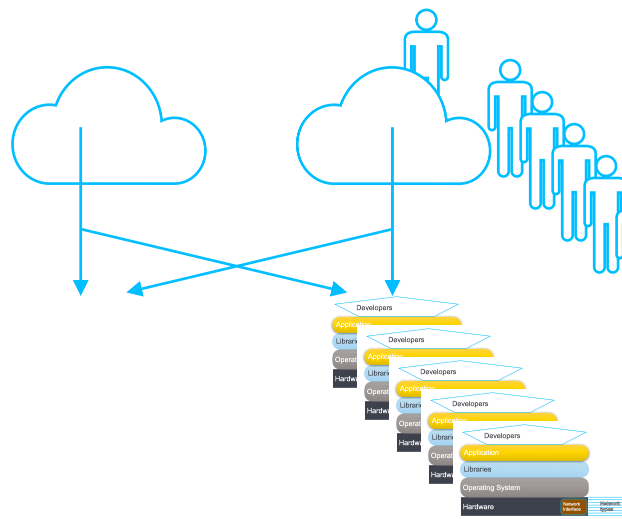
# Distributed Systems Challenges

- Heterogeneity
- Openness
- Security
- Scalability
- Failure handling
- Consistency
- Concurrency
- Transparency
  - Access: *same operations*
  - Location: *any position*
  - Mobility: *dynamic*
  - Replication: *any copy*
  - Failure: *tolerance*
  - Scaling: *any number of nodes*
  - …



# Distributed Systems Challenges

- Heterogeneity
- Openness
- Security
- Scalability
- Failure handling
- Consistency
- Concurrency
- Transparency
  - Access: *same operations*
  - Location: *any position*
  - Mobility: *dynamic*
  - Replication: *any copy*
  - Failure: *tolerance*
  - Scaling: *any number of nodes*
  - …



# Distributed Systems Challenges

- Heterogeneity
- Openness
- Security
- Scalability
- Failure handling
- Consistency
- Concurrency
- Transparency
  - Access: *same operations*
  - Location: *any position*
  - Mobility: *dynamic*
  - Replication: *any copy*
  - Failure: *tolerance*
  - Scaling: *any number of nodes*
  - …